other than liquefied hydrogen.

1.6 In the Code, reference is made to paragraph 5 of the Preamble; paragraph 1.1.6.1; and Note No.8 on completion of certificate in "model form of international certificate of fitness for the carriage of liquefied gases in bulk" in appendix 2 to the Code.

1.7 These Revised Interim Recommendations consist of the following parts. Part A is applicable to ships with any type of cargo containment system. Part B and subsequent part(s) prescribe additional special requirements for cargo containment systems of specific types.

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prescribe additional special requirements for cargo containment systems of specific types.

Part A: General (applicable to ships with any type of cargo containment system);

Part B: Cargo containment systems of independent cargo tanks using vacuum insulation; and

Part C: Cargo containment systems of independent cargo tanks using insulation materials and hydrogen gas in the inner insulation spaces.

1.8 Part A of this document was developed based on the design of parts B and C. If subsequent part(s) are added, the special requirements prescribed in part A may be reviewed.

Part A
General
(Applicable to ships with any type of cargo containment system)

2 GENERAL

2.1 Definition

2.1.1 The following definition should apply for the purpose of these Revised Interim Recommendations.

Permeation is flow of a fluid through another material by diffusion without a defect or

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Recommendations.

Permeation is flow of a fluid through another material by diffusion without a defect or opening of the latter.8

2.2 Requirements for carriage of liquefied hydrogen in bulk

2.2.1 The requirements for the carriage of liquefied hydrogen in bulk have been developed based on the results of a comparison study of similar cargoes listed in chapter 19 of the Code, e.g. liquefied natural gas.

2.2.2 Chapter 19 of the Code governs the application of general requirements for respective cargoes. Selections of the general requirements for respective cargoes are expressed in columns 'c' to 'g'. In addition to general requirements, special requirement s may apply to specific cargoes depending on the properties/hazards of the cargoes.

2.2.3 Tables 1 and 2 specify the proposed selection of the general requirements and the special requirements, respectively, for liquefied hydrogen. In addition to table 2, special

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special requirements, respectively, for liquefied hydrogen. In addition to table 2, special requirements for cargo containment systems of specific types are prescribed in part B or subsequent part(s).

8 1 See paragraph 3.79 of ISO/TR 15916:2015.

Table 1: Requirements for carriage of liquefied hydrogen in bulk

a	b	c	d	e	f	g	h	i
Product name								
Ship type								
Independent tank type C required								
Control of vapour space within cargo tanks								
Vapour detection								
Gauging								
Special requirements								
Hydrogen	2G	-	-	F	C			
See table 2 and, either								

corresponding
table 4 or
table 5, as
appropriate for
the type of
cargo
containment
systems

Table 2: Special requirements for carriage of liquefied hydrogen in bulk

No. Special requirement Related hazard
A-1 Requirements for materials whose design temperature is lower
than -165°C should be agreed with the Administration, paying

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A-1 Requirements for materials whose design temperature is lower
than -165°C should be agreed with the Administration, paying
attention to appropriate standards. Where minimum design
temperature is lower than -196°C, property testing for insulation
materials should be carried out with the appropriate medium, over
a range of temperatures expected in service.
Low temperature
(see 4.2.1)
A-2 Materials of construction and ancillary equipment such as
insulation should be resistant to the effects of high oxygen
concentrations caused by condensation and enrichment at the
low temperatures attained in parts of the cargo system (refer to
the requirement for nitrogen). This special requirement is applied
to all locations where contact with condensed oxygen is
anticipated under normal conditions and foreseeable single
failure scenarios.
Low temperature
(see 4.2.2)
A-3 For cargo pipes containing liquid hydrogen and cold hydrogen
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failure scenarios.
Low temperature
(see 4.2.2)
A-3 For cargo pipes containing liquid hydrogen and cold hydrogen
vapour, measures should be taken to prevent the exposed
surfaces from reaching -183°C. For places where preventive
measures against low temperature are not sufficiently effective,
such as cargo m anifolds, other appropriate measures such as
ventilation which avoids the formation of highly enriched oxygen
and the installation of trays recovering liquid air may be permitted
in lieu of the preventive measures. Insulation on liquid hydrogen

piping syst ems exposing to air should be of non -combustible material and should be designed to have a seal in the outer covering to prevent the condensation of air and subsequent oxygen enrichment within the insulation.
Low temperature
(see 4.2.2)
A-4 Appropriate means, e.g. filtering, should be provided in cargo piping systems to remove impure substances condensed at low temperature.
Low temperature
(see 4.2.3)

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piping systems to remove impure substances condensed at low temperature.
Low temperature
(see 4.2.3)

No. Special requirement Related hazard
A-5 Pressure relief systems should be suitably designed and constructed to prevent blockage due to formation of water or ice.
Low temperature
(see 4.2.4)
A-6 At places where contact with hydrogen is anticipated, suitable materials should be used to prevent any structural deterioration owing to hydrogen embrittlement and degradation of strength and fatigue properties due to continual exposure to hydrogen , as necessary.